FINLITE: A WEB-BASED FINANCIAL TRACKER AND AUTOMATIC REPORT GENERATION SYSTEM WITH AI CHATBOT FOR THE LEAGUE OF INFORMATION TECHNOLOGY ENTHUSIASTS

A Project
Presented to the Faculty of the
College of Computer Studies
Pambayang Dalubhasaan ng Marilao

In Partial Fulfillment
of the requirements for the
System Analysis and Design

By

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1. INTRODUCTION

This chapter presents the first part of the study, introducing the project context by outlining the proposed system, the target organization's background, the business logic and operational processes of the current system, and the specific problems encountered in existing manual operations.

1.1. Project Context

In contemporary organizational management, financial accountability hinges on the systematic recording of monetary flows and the punctual delivery of audit-compliant liquidation reports. A financial tracker serves as a centralized digital mechanism that systematically logs, categorizes, and monitors financial inflows and disbursements in real time, establishing operational transparency, strict custodial accountability, and verifiable audit trails. Complementing this is an automatic report generation system, an automated data processing engine that programmatically compiles, cross-verifies, and formats transactional data into standardized, audit-ready institutional reports, completely removing the vulnerabilities of manual calculation, arithmetic errors, and formatting non-compliance. FinLITE: A Web-Based Financial Tracker and Automatic Report Generation System with AI Chatbot integrates these two core functionalities into an operational financial co-pilot tailored to the unique administrative dynamics of the League of Information Technology Enthusiasts (LITE) at Pambayang Dalubhasaan ng Marilao (PDM).

Pambayang Dalubhasaan ng Marilao (PDM) is a local tertiary institution established pursuant to municipal ordinances to provide accessible, high-quality higher education to the youth of Marilao and surrounding Bulacan municipalities, operating in compliance with the regulatory standards of the Commission on Higher Education (CHED) and the Association of Local Colleges and Universities (ALCU). Within PDM, the College of Computer Studies (CCS) is the primary computing division, offering the flagship Bachelor of Science in Information Technology (BSIT) program under the executive academic leadership of Dean Dr. Emraida Marie M. Manucom and BSIT Program Director Jovylyn Ortiz-Cesar, MBA, MSIT.

The League of Information Technology Enthusiasts (LITE) is the officially accredited academic mother organization representing all students enrolled in the BSIT program at PDM. Guided by Club Advisers Ms. Kimberly Dawn Jatulan and Ms. Krizia Mae Genovia, LITE organizes college-wide technical seminars, programming hackathons, e-sports competitions, social development initiatives, and official

participation in campus events such as Institutional Club Week. LITE is governed by a multi-tiered leadership structure comprising an Executive Board—composed of the President, Vice President, Secretary, Treasurer, and Auditor—elected Year-Level Representatives for the 1st, 2nd, 3rd, and 4th Year cohorts, and nine specialized sub-committees covering Arduino, Cybersecurity, Programming, Networking, Multimedia, Photography, E-sports, Singing, and Dancing disciplines.

LITE operates under severe collegiate physical constraints, possessing no dedicated physical office, private conference room, or institutional workstation. Administrative planning, audit sessions, and ledger consultations take place transiently in campus corridors, student lounges, event booths, or at the adviser's desk located in the second-floor faculty room. Money custody follows strict institutional protocols where all physical cash holdings, petty cash reserves, and accumulated coin collections are secured inside a single small metal cashbox stored inside a locked drawer within the Program Director's Room. The small metal cashbox never leaves the PDM campus. During off-campus purchasing expeditions for event supplies, officers obtain budget allocations directly from the physical cashbox with prior authorization from Club Advisers. Out-of-pocket personal advances (abono) only occur when there is insufficient money inside the cashbox; in such instances, Club Advisers shoulder high-value expenses first, while student executive officers typically shoulder low-value expenses, with all claims queued for reimbursement upon return to campus. Furthermore, leadership transitions occur annually under a strict face-value turnover protocol, where the physical cash in the small metal cashbox must match the exact numerical balance attested on the audited final financial report before outgoing officers receive institutional clearance. Legitimate unpaid reimbursement claims remaining at the end of the academic year are certified and carried over as Accounts Payable liabilities to the incoming administration.

Under certified institutional policy, LITE does not collect mandatory student membership dues. Operating capital originates from four legitimate revenue streams: student entry fees from competitive departmental events and tournaments; batch merchandise orders such as departmental lanyards and organization T-shirts, which strictly enforce one-hundred percent upfront payment prior to vendor ordering; booth rental fees and sales commissions collected from concessionaires during Institutional Club Week; and audited carried-over surplus reserves alongside certified opening liabilities from preceding academic years.

Collection of organizational funds is conducted exclusively through physical cash transactions across collegiate activities. Physical cash collected at event booths by authorized executive officers is subject to a strict holding window, expected by the end of the day or the day after collection to be turned over to the Treasurer for deposit into the

small metal cashbox in the Program Director's room, though in practice this window is frequently delayed during intensive multi-day events. During active event booths, on-duty officers may execute emergency supply deductions directly from unremitted collections, provided an official commercial receipt or signed petty cash acknowledgment slip is submitted immediately upon turnover.

Disbursements for event supplies, certificates, sashes, trophies, and refreshments are governed by organizational authorization protocols. Organizational expenditures are disbursed directly from the physical cashbox; personal out-of-pocket advances, termed Abono, only occur when there are insufficient funds or physical cash in the cashbox. Under this practice, high-value expenditures are shouldered first by Club Advisers, whereas low-value incidental expenses are typically shouldered by student executive officers. These advances represent unliquidated organizational liabilities under Accounts Payable, requiring rigorous tracking of both the original payer and the designated claimant requesting reimbursement.

Liquidation of personal advances requires receipt submission within a strict seven-day grace period following activity completion, although strict enforcement on club advisers is often informal. Commercial purchases require official printed commercial receipts or sales invoices. Non-receipted small cash purchases, such as tricycle fares, public market craft purchases, or emergency printouts, require a handwritten petty cash acknowledgment slip detailing the date, monetary amount, travel route or item purpose, and claimant signature. If approved reimbursement claims exceed available physical cashbox liquidity, a partial cash payout is executed, disbursing available cash while logging the residual balance as an active pending liability.

Physical cash reconciliations are conducted periodically at the adviser's desk in the second-floor faculty room by sorting banknotes and coins from one thousand pesos down to one-peso coins with official witnesses present. The theoretical ledger balance is calculated by adding opening carryover balances and inflows, then subtracting reimbursed disbursements. The discrepancy variance is calculated by subtracting the theoretical balance from total counted physical cash. Balanced counts are certified at zero variance. If a shortage occurs, all organizational spending and club activities immediately freeze until investigated. Untraceable minor shortfalls may be declared as an allowable operating expense labeled Cash Shortage under written justification and dual adviser sign-off, as exemplified by a historical one-hundred sixty-one peso shortage declaration. Cash overages are retained inside the cashbox as unrecorded emergency petty cash buffers.

At semester end, LITE prepares a Final Financial Liquidation Report formatted in Times New Roman typography with borderless tables. Following digital draft review by Club Advisers, hard copies printed on institutional bond paper are routed sequentially across a six-stage wet-ink signatory chain: starting with the Treasurer; followed by the Auditor; moving to the President; proceeding to both Club Advisers for joint dual sign-off; moving to the BSIT Program Director for departmental review; and concluding with the Dean of the College of Computer Studies for final executive approval. If formatting or calculation errors are identified at later stages, the report is rejected, requiring manual re-editing, re-printing, and restarting the physical six-stage routing chain from the beginning.

Primary field interviews with executive officers across successive administrations from 2024 to 2026, corroborated by adviser questionnaires, reveal five critical operational problems within LITE's current manual financial workflows.

The first problem is personal financial vulnerability and delayed reimbursement of personal advances (abono). Because LITE cannot maintain an external operational bank account and must operate out of a single physical cashbox, off-campus purchasing runs require personal out-of-pocket advances whenever cashbox liquidity is insufficient. In such circumstances, Club Advisers are forced to shoulder high-value expenses (such as competition tokens, major event prizes, and large venue materials) first, while student executive officers shoulder low-value supplies and refreshments out-of-pocket. Thermal paper receipts gathered during fast-paced events frequently fade, crumple, or get lost before reaching the campus cashbox. Furthermore, late submission of receipts by student officers past the established seven-day deadline leaves reimbursement queues stalled for weeks, clouding organizational liabilities. Loose receipt slips also lack two-person payment info clarifying who originally paid versus who is claiming reimbursement. When receipts are lost or submitted past deadlines without prior authorization, expenditures cannot be audited, forcing advisers and officers to absorb these institutional costs personally.

The second problem is temporary documentation and administrative leakage in non-receipted small cash purchases. Non-receipted small cash purchases, such as jeepney and tricycle fares, local market craft purchases, and emergency photocopies, are currently documented through loose paper scraps, sticky notes, or unstructured Messenger chat screenshots. These informal records lack standardized information, including exact timestamps, itemized descriptions, authorized claimant identifiers, and supervising officer endorsements. Over a multi-month semester, these paper scraps are smudged or mislaid, depriving the Auditor of verifiable audit trails and resulting in unliquidated expense leakage.

The third problem encompasses physical cashbox custodial risks and high-stress face-value turnover obligations. Storing physical funds in a small metal cashbox inside a locked drawer in the Program Director's room presents custodial exposure as cash is transported across corridors during active campus events. Furthermore, holding collection funds for twenty-four to forty-eight hours creates temporary visibility blind spots. Under the mandatory strict face-value turnover protocol, incoming administrations refuse custody if physical cash deviates by even a single peso from the final audited balance. Because transaction tracking is conducted late on loose paper, historical errors accumulate invisibly, compelling outgoing officers to pay out-of-pocket to rectify balance discrepancies before receiving graduation or leadership clearance.

The fourth problem involves latent discrepancies and slow manual cash computations. Physical cash counts, sorting bills from one thousand pesos down to one-peso bills, are performed weeks or months after events conclude rather than on a daily cadence. In the absence of automated reconciliation tools, officers perform manual cross-multiplication against loose paper tallies, hiding arithmetic errors until final semester report compilation. When cash variances emerge, the resulting mandatory spending and activity freeze paralyzes club operations while officers struggle to determine whether the disparity represents an unlogged petty disbursement, an unremitted collection deduction, or an untraceable coin variance.

The fifth problem consists of institutional liquidation constraints and signatory rejections. Manual preparation of end-of-term liquidation reports using basic word processors results in formatting non-compliance with College of Computer Studies standards, including uneven table borders, misaligned decimal figures, and inconsistent font sizes. Navigating the physical six-stage wet-ink signatory chain, including the mandatory dual adviser joint sign-off, becomes a major delay in the process. If a calculation error, missing voucher attachment, or typographical defect is caught by the Program Director or Dean at the fifth or sixth stage, the report is rejected. Under manual operations, officers must manually re-compute, re-edit, reprint the multi-page document on institutional bond paper, and restart physical wet-ink routing from the first stage, causing immense administrative delay during final examination and graduation periods.

1.2. Objectives of the Study

1.2.1. General Objectives

The general objective of the study is to design and develop FinLITE: A Web-Based Financial Tracker and Automatic Report Generation System with an AI Chatbot for the League of Information Technology Enthusiasts (LITE) at Pambayang

Dalubhasaan ng Marilao to streamline organizational financial transaction tracking, automate institutional liquidation report compilation, and provide intelligent audit assistance.

1.2.2. Specific Objectives

This project has the following specific objectives:

1. To design and implement a secure web-based financial tracker that records cash collections, manages personal out-of-pocket advances (abono) with automated liquidation countdown alerts, and generates electronic petty cash vouchers (e-PCVs) to prevent unliquidated expense leakages.
2. To develop an interactive cash denomination counter and automated variance reconciliation engine that tallies physical bills and coins against single-entry ledger balances to eliminate arithmetic errors, record cash shortage justifications, and monitor custodial turnovers.
3. To deploy a one-click institutional document generation engine that compiles CCS-compliant liquidation reports in editable Word (.docx) and PDF formats, integrated with an AI Chatbot co-pilot that provides instant budget and expense inquiries for executive officers and club advisers.

1.3. Purpose and Description

This study is significant to the following:

1.3.1 LITE Organization Student Officials:

The system benefits student officials by replacing paper records with a digital ledger, ensuring fast reimbursement of personal advances (abono), and protecting outgoing officers from paying cash shortages out-of-pocket during turnover.

1.3.2 Club Advisers:

The system benefits advisers by giving them real-time visibility over club funds, streamlining the digital review and approval of reimbursement claims, and allowing them to ask the AI Chatbot for instant budget updates.

1.3.3 BSIT Students:

The system will benefit the general student body of the Bachelor of Science in Information Technology program by upholding financial integrity and accountability in the management of student activity fees, merchandise payments, and event booth

revenues, thereby ensuring that organizational funds are utilized efficiently for meaningful collegiate technical activities and student development.

1.3.4 BSIT College Leadership:

The system benefits the Program Director and the Dean by ensuring all submitted financial reports follow correct CCS formatting and math, avoiding repeated document rejections and delays.

1.3.5 The Proponents:

The system benefits the researchers by helping them apply their skills in web programming, database design, and AI integration to solve a real campus problem.

1.3.6 Future Researchers:

The system benefits future researchers by serving as a useful guide and reference for building financial tools for student organizations.

1.4. Scope and Delimitation

This study focuses on the design and development of FinLITE, a web-based financial tracker and report generation system with an AI chatbot for the League of Information Technology Enthusiasts (LITE) at Pambayang Dalubhasaan ng Marilao (PDM). The system starts with user authentication and is divided into specific modules aimed at replacing the organization's manual record-keeping and solving common financial tracking issues.

The system workflow initiates with the User Authentication and Role-Based Access Control Module, where the LITE president and vice president, treasurer, auditor, and faculty club advisers securely register and log in to the web platform using validated @gmail.com accounts via Google OAuth 2.0 or email and password. Upon credential verification and approval at the administrative waiting gate, the club advisers become the system administrators and assign distinct operational permissions according to designated organizational roles. The Treasurer gains permissions to encode cash collections and upfront merchandise payments into the ledger, record advance reimbursement payouts from the physical cashbox, operate the digital denomination keypad during physical cash counts, and generate one-click CCS-compliant liquidation reports. The Auditor receives read-only verification access to inspect uploaded receipt photos and vouchers without altering records, monitor missing documentation in real time, review automated variance flags, record approved shortage justifications or cashbox buffers, and execute digital audit

sign-offs. Meanwhile, the LITE President and Vice President obtain rights to track real-time organizational budget balances and outstanding debts, preview and edit report text directly in-browser, execute executive digital sign-offs, and unlock returned reports for quick re-export. The Faculty Club Advisers maintain administrative oversight, accessing real-time dashboard monitoring of liquid assets and pending debts, digital approval controls for expenditures exceeding the one thousand peso threshold, joint dual sign-off capabilities for finalized reports, and natural-language inquiry access to the AI Chatbot co-pilot. Additionally, authorized executive officers hold permissions to log event booth collection remittances under custodial tracking and generate electronic petty cash vouchers with preset travel routes for official campus errands.

Once logged in, the Club Adviser and executive officers access the Advance & Reimbursement Management Module to resolve personal financial vulnerability and delayed reimbursement of out-of-pocket advances (abono). In accordance with organizational rules, purchases are primarily funded directly from the physical cashbox; personal advances only occur when there are insufficient funds or physical liquidity in the cashbox. When funds are depleted, Club Advisers shoulder high-value expenses (such as competition tokens, major event prizes, and large venue materials) first, while student executive officers shoulder low-value supplies and refreshments out-of-pocket. When an officer or Club Adviser advances personal funds, they open this module on their smartphone or computer to file an immediate claim. Using a dual-entity tracking form, the claimant designates both the original payer and the designated claimant, then uploads a photo of the commercial sales invoice. The module automatically initiates a seven-day liquidation countdown timer with visual status badges (Active, Warning, Overdue) before thermal receipt ink fades or slips get lost. Through a restricted interface, the Auditor performs real-time receipt tracking and read-only verification to inspect the uploaded proofs and catch missing claimant details early, while the Club Advisers execute digital approval of the abono claim. Meanwhile, the LITE President or Vice President monitors the outstanding reimbursement debt on the budget dashboard, and the Treasurer accesses the abono payout queue to execute cashbox reimbursements, systematically logging full or partial payout states and updating creditor balances to ensure no leader absorbs organizational expenses out of pocket.

To address the second problem of temporary documentation and administrative leakage in non-receipted micro-disbursements, the authorized purchasing officer—such as the LITE President, Vice President, or Treasurer—accesses the Electronic Petty Cash Voucher (e-PCV) Generator embedded within the unified transaction recording interface. An Electronic Petty Cash Voucher (e-PCV) is a standardized digital disbursement form used to substantiate, calculate, and route out-of-pocket micro-expenses and non-receipted transit fares without fragile paper slips. For low-value transit and incidental expenses that

lack official commercial receipts, such as tricycle and jeepney fares between PDM and Marilao Poblacion, emergency photocopies, or public market craft purchases, the purchasing officer generates a digital voucher directly on screen. The officer selects from standardized travel route presets or expense categories, receives an auto-indexed voucher tracking number, inputs the itemized description, and captures their on-screen digital claimant signature alongside the supervising digital endorsement of the LITE President or Vice President. Once submitted, the system attaches the generated voucher directly to the transaction ledger; the Auditor accesses the restricted interface to perform read-only verification, inspecting the voucher details, appending audit remarks, and tracking claimant metadata in real time to prevent expense leakage, while the Treasurer records the corresponding cashbox disbursement, permanently eliminating fragile paper scraps, sticky notes, and unstructured chat screenshots.

The Treasurer and LITE President or Vice President then transition to the Witnessed Custodial Tracking and Turnover Module to resolve physical cashbox custodial risks and high-stress face-value turnover obligations. When event collections conclude, the system activates an automated holding-window timer that monitors physical funds held by on-duty collecting officers, dispatching digital alerts to the Treasurer, LITE President or Vice President, and Club Advisers if collections remain unremitted past the required 24-to-48-hour window before being locked in the metal cashbox in the Program Director’s Room. Upon remittance, the Treasurer encodes the turned-over cash into the single-entry ledger. At the conclusion of an academic tenure, the outgoing Treasurer and LITE President and Vice President run the turnover protocol witnessed by the Auditor and Club Advisers, generating an official Turnover Reconciliation Certificate that validates counted physical cash against audited ledger balances down to the exact peso. The module formally archives legitimate unsettled reimbursement claims as certified Accounts Payable carryover liabilities for the incoming administration, safeguarding outgoing leaders from unjustified out-of-pocket balance deductions during graduation clearance.

To eliminate latent arithmetic errors and prevent paralyzing club-wide spending freezes, the Treasurer and Auditor access the Interactive Cash Denomination Counter and Variance Reconciliation Engine during periodic physical audits conducted at the adviser's desk. Instead of computing loose paper tallies by hand, the Treasurer inputs the physical quantity of each bill (₱1,000, ₱500, ₱200, ₱100, ₱50, ₱20) and coin denomination (₱20, ₱10, ₱5, ₱1) into the digital keypad calculator. The engine instantly computes the total cash on hand and cross-references it in real time against the theoretical ledger balance, triggering an automated variance alert for the Auditor if a discrepancy is detected. Under the discrepancy logging protocol, if a verified untraceable minor shortfall occurs, the system automatically drafts a standardized Cash Shortage Justification Report for the

Auditor to log and the Club Advisers to endorse through digital dual sign-off. Conversely, any cash overage is credited directly to the emergency petty cash buffer, maintaining transparent accountability while preventing unwarranted operational freezes.

Finally, the Treasurer and LITE President or Vice President navigate to the one-click Institutional Document Generation Engine and AI Chatbot Co-Pilot to solve institutional liquidation bottlenecks, formatting non-compliance, and signatory rejections. With a single click, the Treasurer triggers the export engine, which pulls all validated ledger entries, electronic petty cash vouchers (e-PCVs), and reconciliation logs to programmatically compile end-of-term financial reports in Microsoft Word (.docx) and PDF formats. These reports strictly comply with College of Computer Studies (CCS) standards, featuring Times New Roman typography, 12pt font sizing, borderless tables, right-aligned two-decimal currency figures, and the pre-formatted six-stage wet-ink signatory block. Directly in-browser, the LITE President or Vice President previews the exact report layout to make live text and metadata adjustments before export, and possesses the capability to unlock returned reports for quick re-export without re-typing the entire document from scratch. The report then progresses through the system's digital routing workflow, securing online certification from the Auditor at Stage 2, executive approval from the LITE President or Vice President at Stage 3, and digital dual sign-off from both Club Advisers at Stage 4 prior to physical printing for the Program Director and Dean. Concurrently, an embedded zero-hallucination AI Chatbot Co-Pilot is accessible across the platform, allowing executive officers and Club Advisers to submit natural-language inquiries to check remaining event budgets, monitor reimbursement debts, and inspect category totals instantly without manual spreadsheet combing.

The operational boundaries of this study are defined by several delimitations and governance safeguards to preserve institutional compliance and feasibility. First, FinLITE operates strictly as a physical cash ledger, denomination counter, and audit tracking system, and does not integrate digital financial payment gateways, mobile wallet interfaces (such as GCash or Maya), electronic fund disbursements, or banking APIs; all recorded revenues originate from physical cash collections, booth receipts, and upfront merchandise payments. Second, the system does not encompass automated physical currency handling, mechanical coin sorting, or electronic locking mechanisms for the physical metal cashbox stored inside the Program Director's room, operating strictly as a digital ledger, calculation, and audit tracker. Third, to preserve unbroken audit integrity across academic tenures, the platform enforces a Financial Audit Preservation Rule that prohibits the deletion of user accounts linked to existing transactions, cash reconciliations, or audit remarks, restricting administrative action strictly to account suspension. Fourth, API rate-limiting middleware restricts request volumes across authentication, report generation, and AI chat endpoints to protect server stability, while

outbound email dispatches (such as registration alerts and password reset tokens) are currently simulated in developer consoles pending live institutional SMTP deployment.

Fifth, while the platform automates report calculation, layout formatting, and pre-audit validation, it does not bypass or replace the physical wet-ink signatory routing chain required by collegiate governance; final generated reports must still be printed on institutional bond paper and manually signed across all six administrative stages for official institutional filing. Lastly, the system's workflow rules, approval protocols, and reporting templates are customized exclusively for the League of Information Technology Enthusiasts (LITE) under the College of Computer Studies at Pambayang Dalubhasaan ng Marilao, and do not apply to other academic organizations, campus student councils, or external academic departments.

2. TECHNICAL BACKGROUND OF THE STUDY

2.1 Project Analysis

The analysis phase thoroughly investigated the existing financial tracking, cash custody, and liquidation routines of LITE across successive administrations from 2024 to 2026. Data gathered through questionnaires via Google Forms with the 2026-2027 LITE president and club advisers revealed critical inefficiencies in receipt handling, personal mobile wallet commingling, arithmetic reconciliation, and multi-stage hardcopy routing. To establish a rigorous architectural foundation, the operational workflows of both the existing manual system and the proposed FinLITE platform were formally modeled using structured Data Flow Diagrams (DFDs) following standard Gane-Sarson notation.

2.1.1 Existing Data Flow Diagram

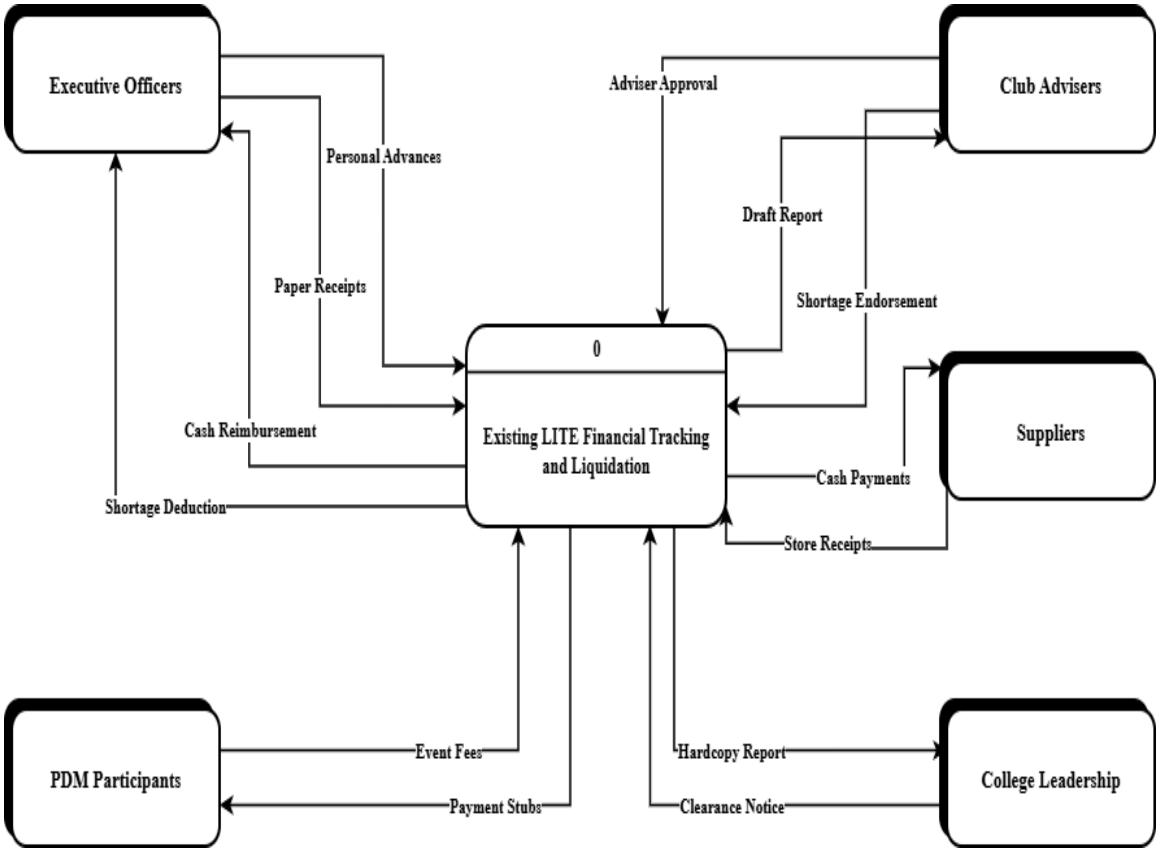


Figure 1: Context Diagram of the Existing LITE Financial Tracking and Liquidation

Figure 1 illustrates the Context Diagram of LITE Financial Tracking and Liquidation, showing the high-level boundary and external information exchanges of LITE's existing manual financial workflow. In the current system, financial operations are driven by physical cash exchanges, loose paper documentation, and manual verbal or chat requests. When officers or club advisers execute emergency purchasing runs for competition materials, trophies, and food, they front personal capital (Abono) and gather thermal paper receipts, which are manually handed over to the Treasurer. Students and tournament participants hand over physical cash entry fees to on-duty officers, receiving handwritten paper acknowledgment slips or event wristbands in return.

Physical cash holdings are held by student collectors during events and later deposited into a single small metal cashbox secured inside a locked drawer in the BSIT Program Director's Room. Expenditures exceeding one thousand pesos require written or chat endorsement from both Club Advisers. At the conclusion of the academic semester, the Treasurer manually formats a multi-page liquidation report using desktop word processors, which must navigate a physical, sequential six-stage wet-ink signatory chain

(Treasurer, Auditor, President, Dual Club Advisers, Program Director, and CCS Dean). Any calculation or typographical error detected by college administrators results in complete rejection, forcing manual re-editing, re-printing on bond paper, and restarting the entire six-stage routing chain from the beginning

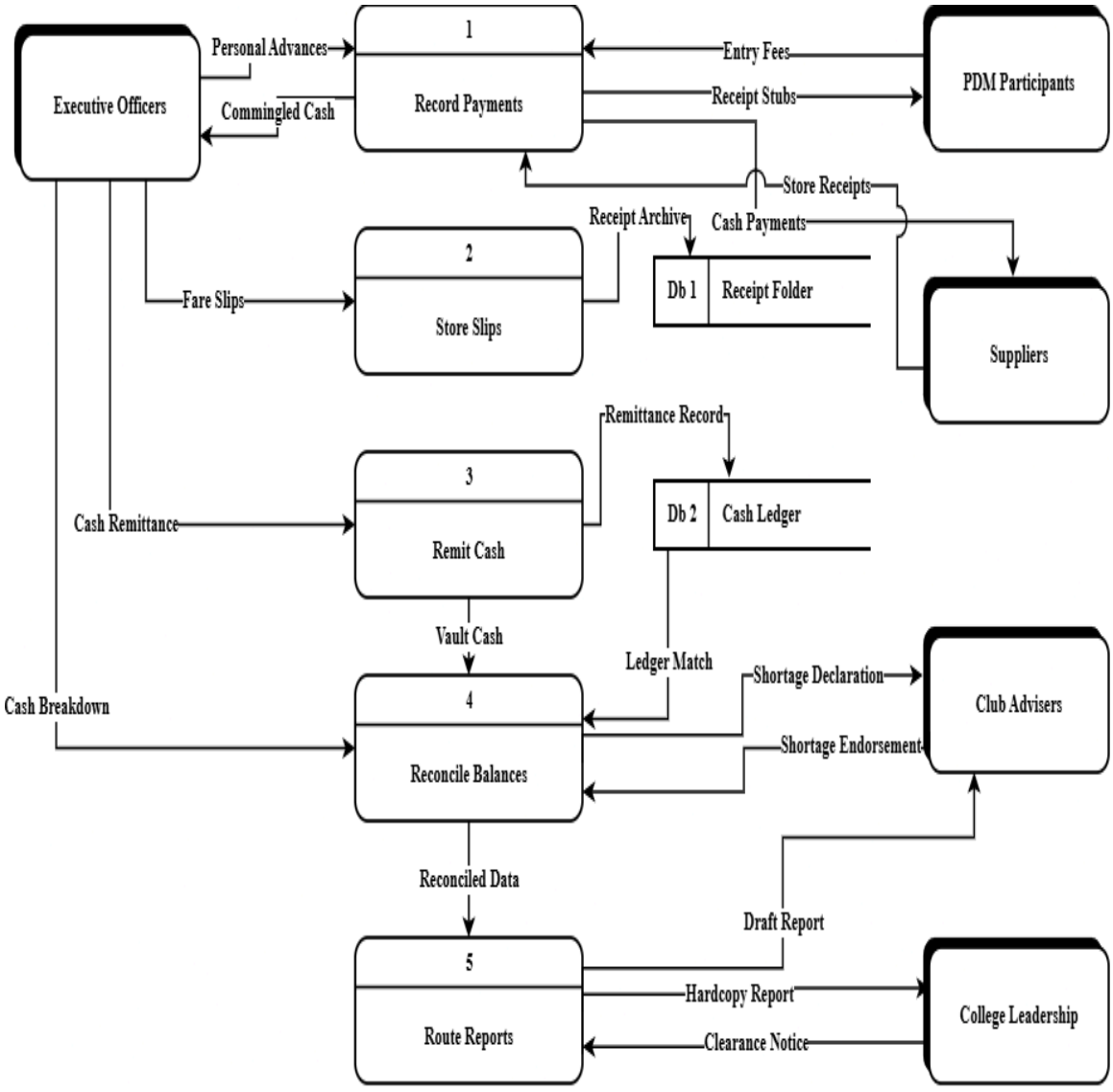


Figure 2: Existing Diagram 0 of LITE Financial Tracking and Liquidation

Figure 2 shows that whenever the student organization prepares for an event, executive officers usually use their own personal money first to buy supplies from local stores and vendors. In return, the suppliers hand them printed thermal receipts, which the officers keep inside a paper folder. At the same time, students participating in the event pay their registration fees in cash and receive a simple tear-off paper stub as proof of

payment. Because officers collect this cash on the go, the club money often mixes with their personal funds in their pockets and wallets, which they later use to pay themselves back for what they spent in advance. For travel expenses like jeepney and tricycle rides where drivers cannot issue official receipts, the officers write down their fare details on handwritten slips and store them in the folder alongside the other receipts.

Once collections and errands are done, the officers hand over the collected cash to the treasurer. The treasurer notes down who is holding the funds, writes the remittance into a physical ledger notebook, and places the physical bills and coins into storage. When it is time to check the funds, the treasurer pulls out the stored cash and manually counts every bill and coin. This physical count is compared with the previous balance recorded in the ledger to see if the amounts match. If any money is missing, the officers have to write an explanation letter and submit it to their club advisers to get a signed endorsement. Finally, the treasurer prepares a draft report, gathers all the paper receipts from the folder, matches them with the records in the ledger, and prints out a final hardcopy report to submit to college leadership for institutional review and clearance

2.1.2 Proposed Data Flow Diagram

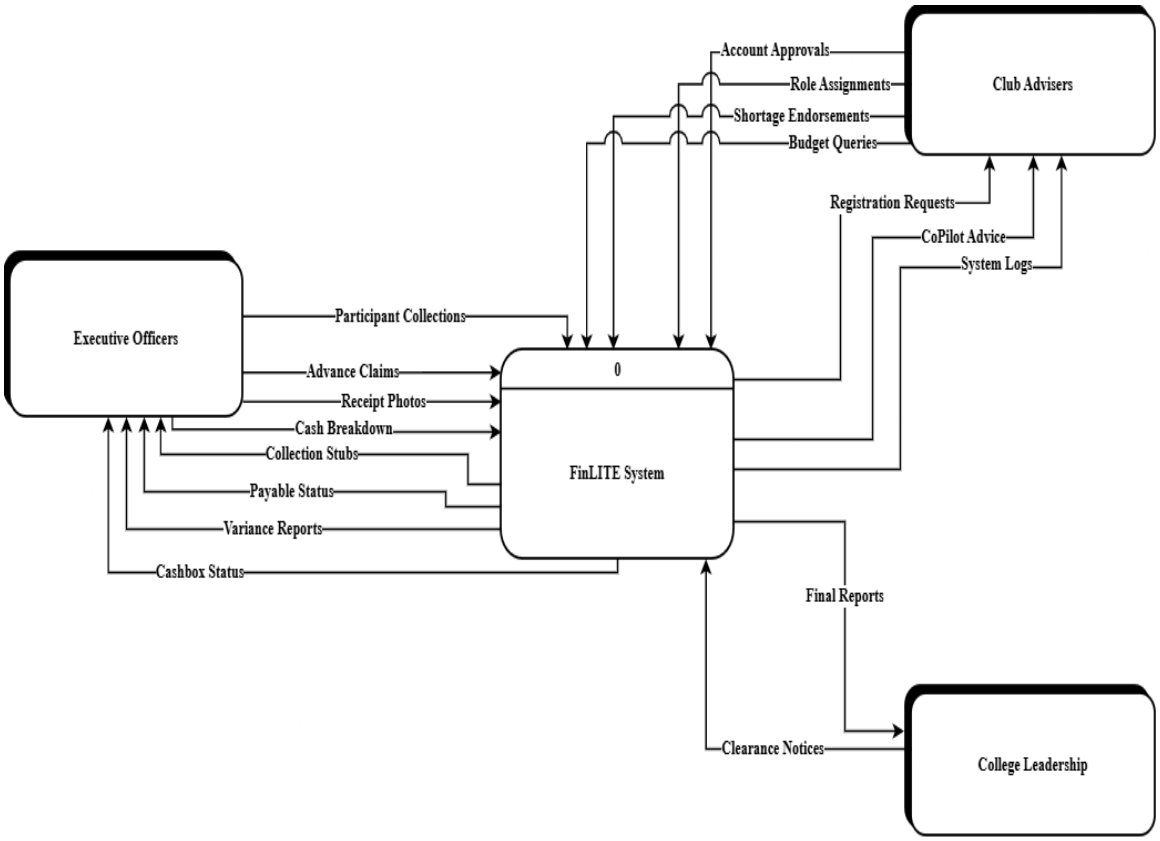


Figure 3: Context Diagram of FinLITE System

Figure 3 illustrates the Context Diagram (Level 0) of the proposed FinLITE financial management system, adhering strictly to the Gane and Sarson notation. FinLITE is represented as a singular top-level process, 0.0 Manage Finances, interfacing with three primary external institutional entities: Executive Officers, Club Advisers, and College Leadership.

Operational custody of the organization's physical cashbox resides with the Executive Officers (Treasurer, Auditor, and Purchasing Officers). Officers supply operational inflows including Participant Collections received at event booths, Advance Claims for authorized personal reimbursements, digital Receipt Photos, and denomination-specific Cash Breakdown figures recorded during physical cashbox counts. In response, FinLITE returns verified Collection Stubs, real-time Payable Status, automated Variance Reports, and current Cashbox Status.

Institutional governance and system administration are consolidated under the Club Advisers, who simultaneously serve as academic mentors and system administrators. Advisers govern user onboarding via Account Approvals and Role Assignments, authorize high-value disbursements through Adviser Approvals exceeding the ₱1,000 organizational ceiling, submit Shortage Endorsements, and execute Budget Queries. FinLITE provides advisers with incoming Registration Requests, proactive Threshold Alerts, heuristic CoPilot Advice, and immutable System Logs tracking all financial events.

Finally, the College Leadership (BSIT Program Director and CCS Dean) represents the overarching institutional clearance authority. At the end of each academic term, FinLITE compiles comprehensive Final Reports (liquidation sheets, cashbox reconciliation certificates, and audit trails), prompting the issuance of institutional Clearance Notices necessary for organizational accreditation.

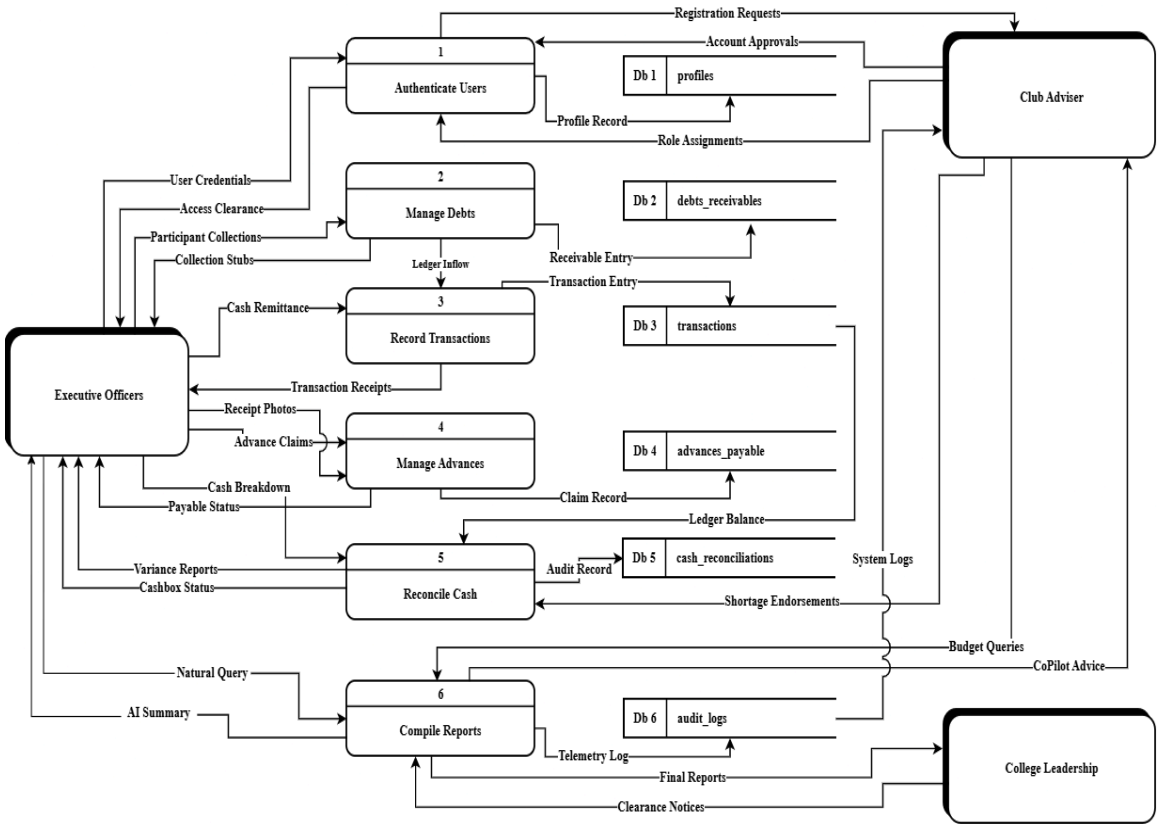


Figure 4: Diagram 0 of FinLITE System

Figure 4 presents the proposed FinLITE platform, decomposing the top-level process into six discrete functional processes supported by six persistent relational data stores while preserving balanced boundary exchanges with the external entities. In Process 1 (Authenticate Users), applicant officers submit login credentials through either Google OAuth 2.0 or standard credentials that undergo automated @gmail.com syntax validation and typo filtering, placing unapproved accounts into a quarantined waiting view while routing Registration Requests to the Faculty Club Advisers, who issue administrative Account Approvals and Role Assignments that commit active privileges to Db 1 (profiles) and return cryptographic Access Clearance to the authenticated Executive Officers. In Process 2 (Manage Debts), collecting officers encode merchandise pre-orders and installment collections as Participant Collections, persisting debtor obligations into Db 2 (debts_receivables) and issuing electronic Collection Stubs, wherein fully settled installment contracts automatically route as Ledger Inflow into Process 3 (Record Transactions). In Process 3.0 (Record Transactions), the Treasurer inputs operational income as Cash Remittance and records expenditures with attached Receipt Photos, checking available cashbox funds to prevent overdrafts before

committing ledger entries to Db 3 (transactions) and generating verified Transaction Receipts.

In Process 4 (Manage Advances), officers who fronted personal funds submit out-of-pocket Advance Claims and receipt proofs into Db 4 (advances_payable), where the system initiates a 7-day post-activity liquidation countdown, enabling the Treasurer to verify receipt authenticity and disburse full or partial cash reimbursements directly from the cashbox while returning updated Payable Status to the claimant. In Process 5 (Reconcile Cash), the Treasurer inputs piece counts across all 11 legal tender Philippine denominations as Cash Breakdown, saving an immutable Audit Record into Db 5 (cash_reconciliations) and cross-referencing counted physical currency against the theoretical Ledger Balance fetched from Db 3 (transactions) to generate Variance Reports and live Cashbox Status, with verified discrepancies submitted to the Club Advisers for Shortage Endorsements. Finally, in Process 6 (Compile Reports), operational telemetry is recorded as Telemetry Log entries in Db 6 (audit_logs) while the engine aggregates verified data across Db 3, Db 4, and Db 5, enabling Executive Officers to dispatch conversational Natural Query prompts that return grounded AI Summary insights for strategic liquidity management, while concurrently allowing Club Advisers to submit institutional Budget Queries that yield grounded CoPilot Advice and inspect historical System Logs from Db 6, culminating in the compilation of comprehensive Final Reports transmitted to College Leadership to secure formal Clearance Notices.

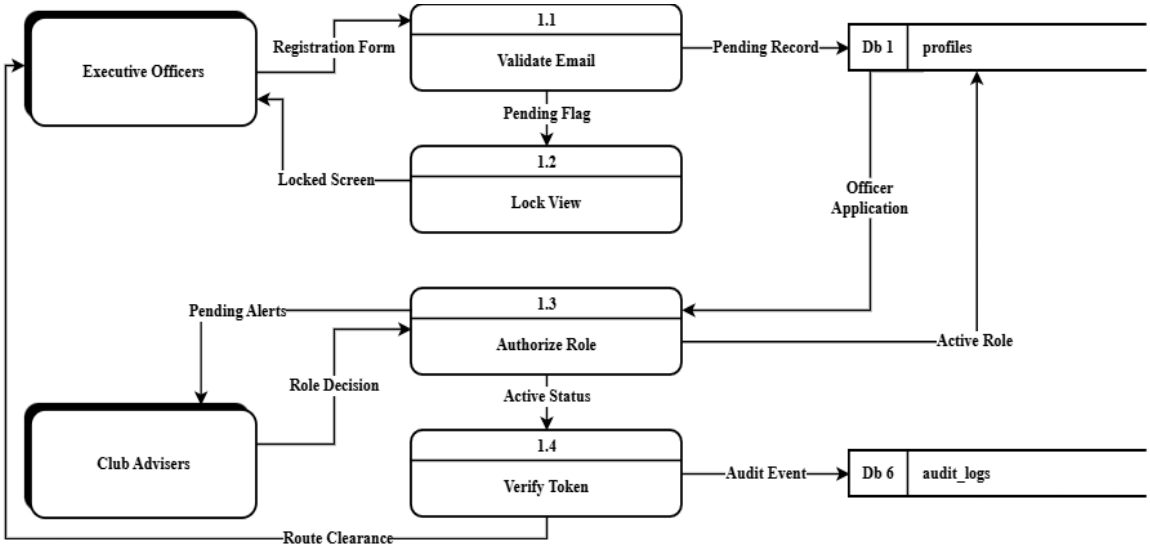


Figure 5: Diagram 1 of Process 1: Authenticate Users

Figure 5 illustrates the Level 2 decomposition of Process 1.0 (Authenticate Users) across four operational sub-processes. When an applicant officer submits a Registration Form, Sub-process 1.1 (Validate Email) validates user credentials and writes a pending profile to Db 1 (profiles). Sub-process 1.2 (Lock View) places unapproved accounts in a quarantined state, presenting a Locked Screen to the applicant while issuing pending alerts to the Club Advisers. Upon adviser review, Sub-process 1.3 (Authorize Role) receives Role Decision directives from the adviser, updating user privileges in Db 1 to active status. Finally, Sub-process 1.4 (Verify Token) issues cryptographic Route Clearance session tokens to the authenticated officer while logging an immutable Audit Event to Db 6 (audit_logs).

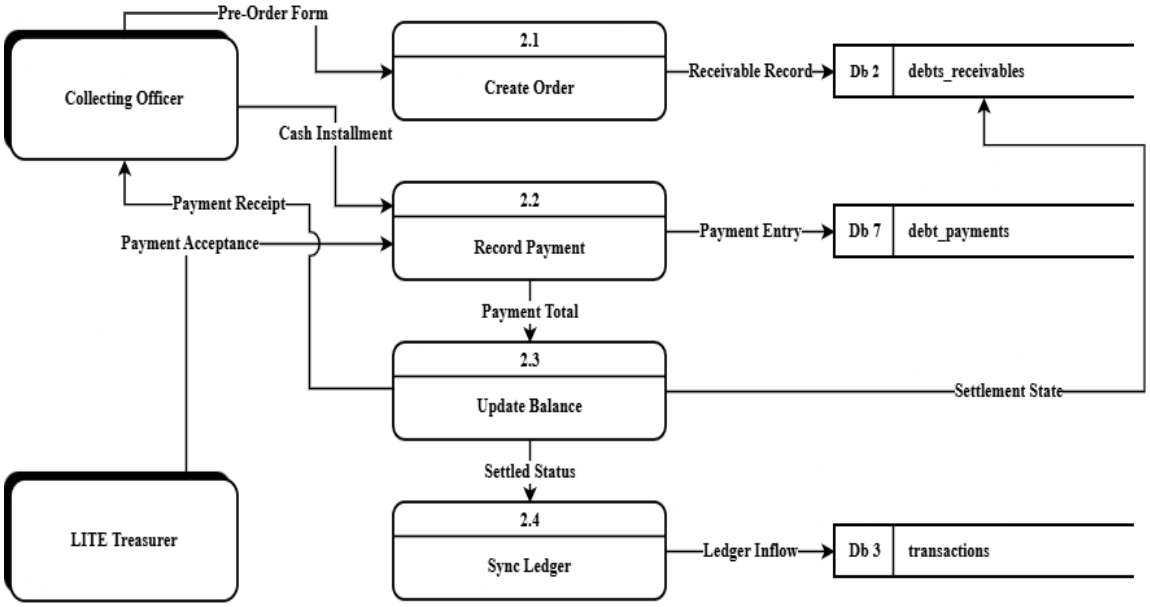


Figure 6: Diagram 1 of Process 2: Manage Debts

Figure 6 outlines the Level 2 decomposition of Process 2 (Manage Debts) across four sub-processes that manage merchandise pre-orders and installment commitments (hulog).

In Sub-process 2.1 (Create Order), a collecting officer submits a Pre-Order Form containing the student buyer's name, student number, contact details, and merchandise specifications (departmental lanyards, T-shirts), creating an active contract profile (Receivable Record) in Db 2 (debts_receivables). When the student remits cash at an event booth, Sub-process 2.2 (Record Payment) receives the Cash Installment

countersigned by the Treasurer's Payment Acceptance, recording each partial payment into Db 7 (debt_payments).

Sub-process 2.2 relays the Payment Total to Sub-process 2.3 (Update Balance), which dynamically computes cumulative installment payments against total contract obligations, updating the Settlement State in Db 2 across defined lifecycle states (Unpaid, Partially Paid, Settled) and generating an electronic Payment Receipt for the collecting officer to issue to the participant. Upon achieving full settlement (Settled), Sub-process 2.3 passes a Settled Status trigger to Sub-process 2.4 (Sync Ledger), which automatically generates a verified Ledger Inflow in Db 3 (transactions), guaranteeing arithmetic parity between receivables and cashbox holdings without redundant double-entry.

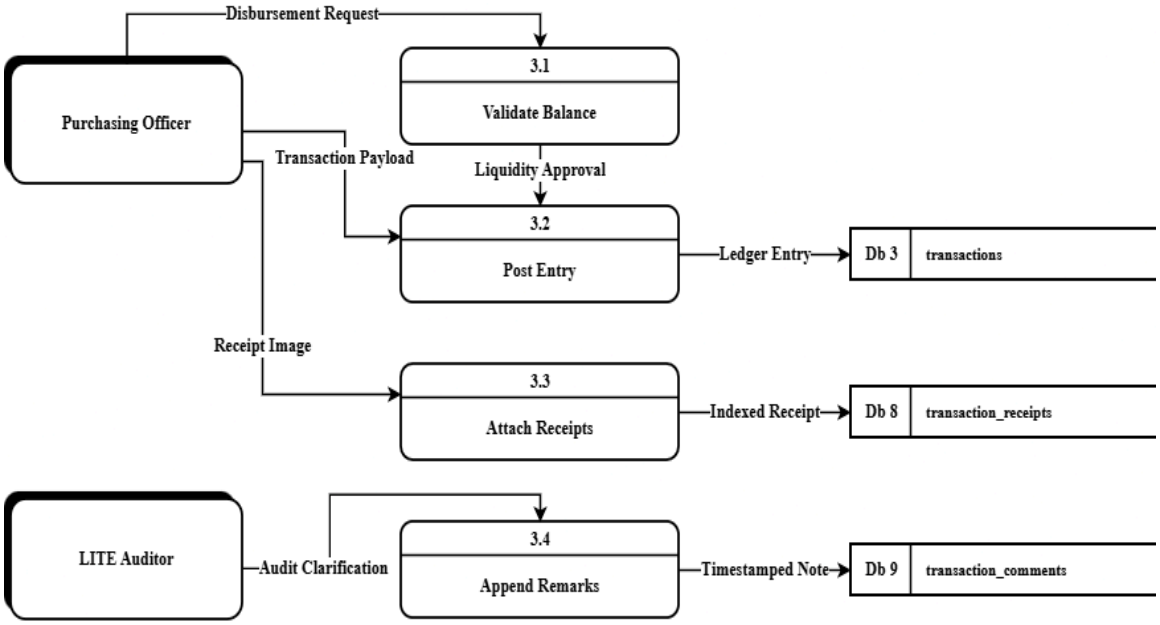


Figure 7: Diagram 1 of Process 3: Record Transactions

Figure 7 details the internal mechanisms of Process 3 (Record Transactions) across four specialized sub-processes governing financial entries, image proof indexing, and auditor collaboration.

When an operational expenditure is proposed, Sub-process 3.1 (Validate Balance) receives a Disbursement Request from the Purchasing Officer and evaluates requested funds against liquid currency in the cashbox; if funds are sufficient, it issues Liquidity Approval to Sub-process 3.2 (Post Entry). Concurrently, the purchasing officer submits

the operational Transaction Payload (amount, date, category, narrative remarks), enabling Sub-process 3.2 to write a verified Ledger Entry to Db 3 (transactions).

To satisfy collegiate evidentiary standards, Sub-process 3.3 (Attach Receipts) captures uploaded Receipt Image proofs (commercial invoices or handwritten petty cash slips for tricycle fares), applies client-side compression and canvas cropping, and commits the Indexed Receipt into Db 8 (transaction_receipts). To facilitate transparent post-audit reviews without altering verified figures, Sub-process 3.4 (Append Remarks) allows the LITE Auditor to submit an Audit Clarification, persisting a Timestamped Note into Db 9 (transaction_comments).

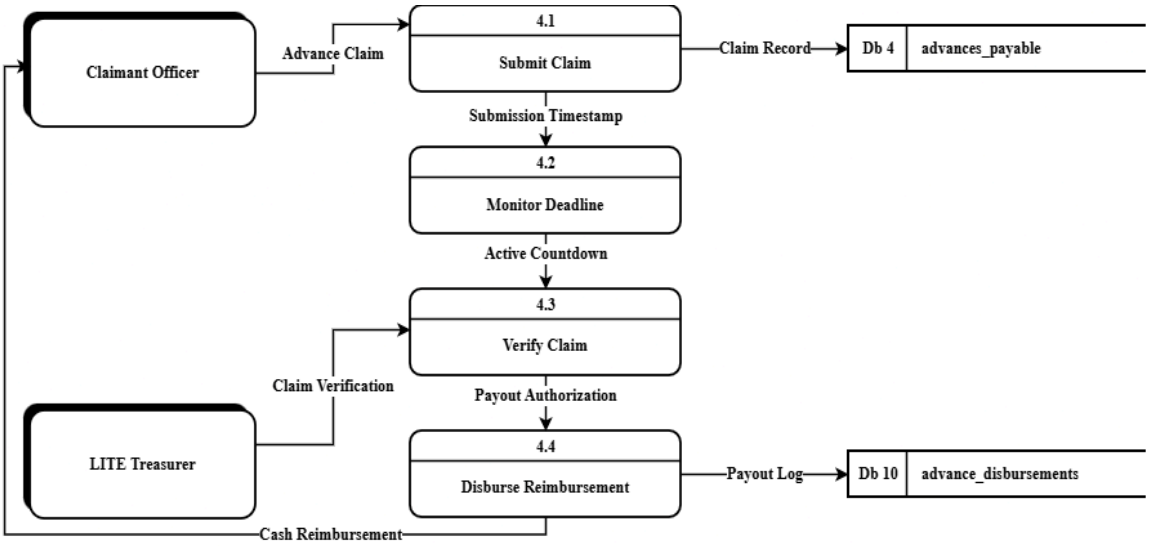


Figure 8: Diagram 1 of Process 4: Manage Advances

Figure 8 illustrates the operational logic of Process 4 (Manage Advances), governing out-of-pocket organizational expenses (abono) from claim submission to cash reimbursement.

In Sub-process 4.1 (Submit Claim), an officer who fronted personal capital submits an Advance Claim accompanied by digital receipt proofs, creating an accounts payable liability record (Claim Record) in Db 4 (advances_payable). Sub-process 4.1 transmits the Submission Timestamp to Sub-process 4.2 (Monitor Deadline), which initiates an automated Active Countdown tracking the institutional 7-day post-activity liquidation window, updating visual status badges (Active, Warning, Overdue).

In Sub-process 4.3 (Verify Claim), the LITE Treasurer reviews the digitized proofs, verifies receipt authenticity, and issues Claim Verification, generating immediate

Payout Authorization to Sub-process 4.4 (Disburse Reimbursement). If liquid cash in the box is temporarily constrained, Sub-process 4.4 supports partial installment payouts, recording each disbursement into Db 10 (advance_disbursements) as a Payout Log and issuing physical Cash Reimbursement directly to the Executive Officers (Claimant Officer), systematically decrementing the residual creditor liability in Db 4 until the advance is fully settled.

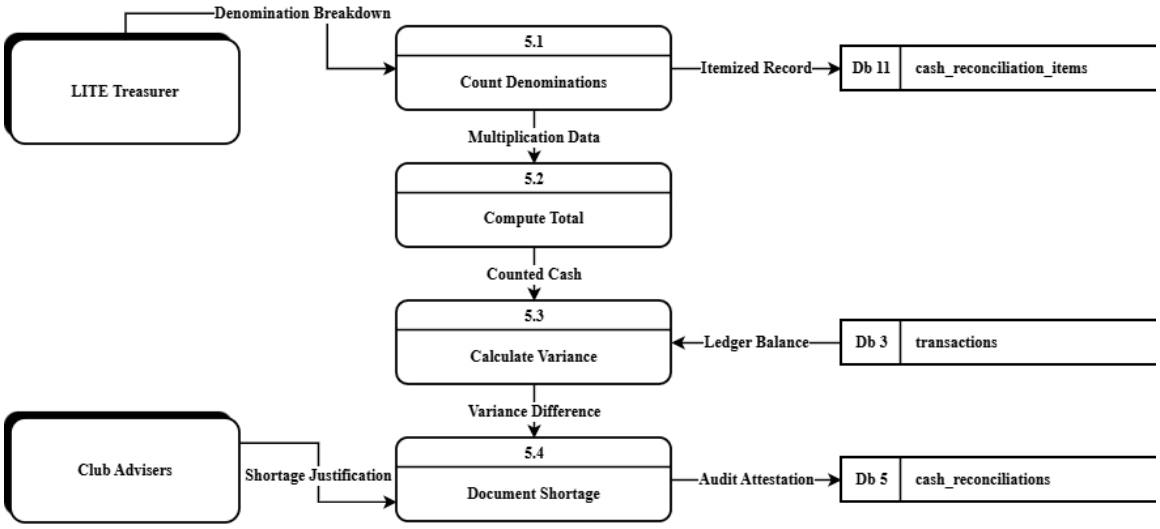


Figure 9: Diagram 1 of Process 5: Reconcile Cash

Figure 9 presents the architectural breakdown of Process 5 (Reconcile Cash), coordinating physical currency audits conducted at the adviser's desk inside the Program Director's Room.

In Sub-process 5.1 (Count Denominations), the Executive Officers (LITE Treasurer) physically counts cashbox currency and inputs piece-counts across all 11 legal tender Philippine denominations (banknotes: ₱1,000, ₱500, ₱200, ₱100, ₱50, ₱20; coins: ₱20, ₱10, ₱5, ₱1, and ₱0.25 centavos) as Denomination Breakdown, persisting itemized counts into Db 11 (cash_reconciliation_items). Sub-process 5.1 forwards Multiplication Data to Sub-process 5.2 (Compute Total), which computes the exact physical currency subtotal with integer-centavo precision, supplying Counted Cash figures to Sub-process 5.3 (Calculate Variance).

Sub-process 5.3 pulls the theoretical balance (Ledger Balance) from Db 3 (transactions) and computes the variance (Counted Physical Cash minus Theoretical Balance). If an arithmetic discrepancy occurs, the Variance Difference is routed to

Sub-process 5.4 (Document Shortage), where an official Shortage Justification is drafted and endorsed by the Club Advisers (Adviser & Admin), committing a certified Audit Attestation into Db 5 (cash_reconciliations) under dual-officer attestation.

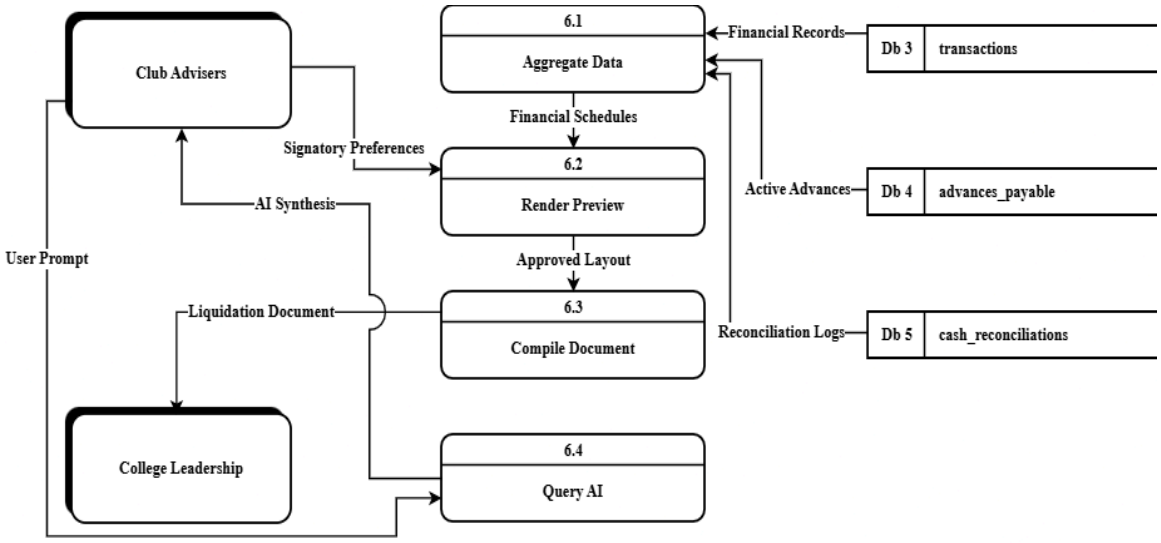


Figure 10: Diagram 1 of Process 6: Compile Reports

Figure 10 outlines the Level 2 mechanisms of Process 6 (Compile Reports), managing automated liquidation synthesis and natural-language financial inquiry.

In Sub-process 6.1 (Aggregate Data), the reporting engine extracts verified financial datasets: Financial Records from Db 3 (transactions), Active Advances from Db 4 (advances_payable), and Reconciliation Logs from Db 5 (cash_reconciliations), compiling scoped Financial Schedules. Sub-process 6.2 (Render Preview) generates a pixel-accurate 1:1 live preview canvas incorporating adviser Signatory Preferences (toggling required administrative signatures across Treasurer, Auditor, Advisers, Program Director, and Dean).

Upon confirmation, Sub-process 6.3 (Compile Document) exports the verified figures into a standardized Microsoft Word (.docx) liquidation packet (Liquidation Document) featuring Calibri 11pt typography, borderless financial tables, formal transmittal headers to Co-Curricular Affairs Coordinator Ms. Ligaya H. Estrella, and automated 2-column, 4-up receipt attachment sheets, transmitting the packet to College Leadership for institutional clearance. Concurrently, Sub-Process 6.4 executes conversational intelligence queries. Authorized officers—including the President, Vice

President, Treasurer, and Club Advisers—dispatch natural language prompts into the conversational co-pilot, prompting the system to retrieve live database figures, calculate net liquidity positions, and return grounded financial syntheses without risk of hallucination.

2.1.3 Use Case Diagram

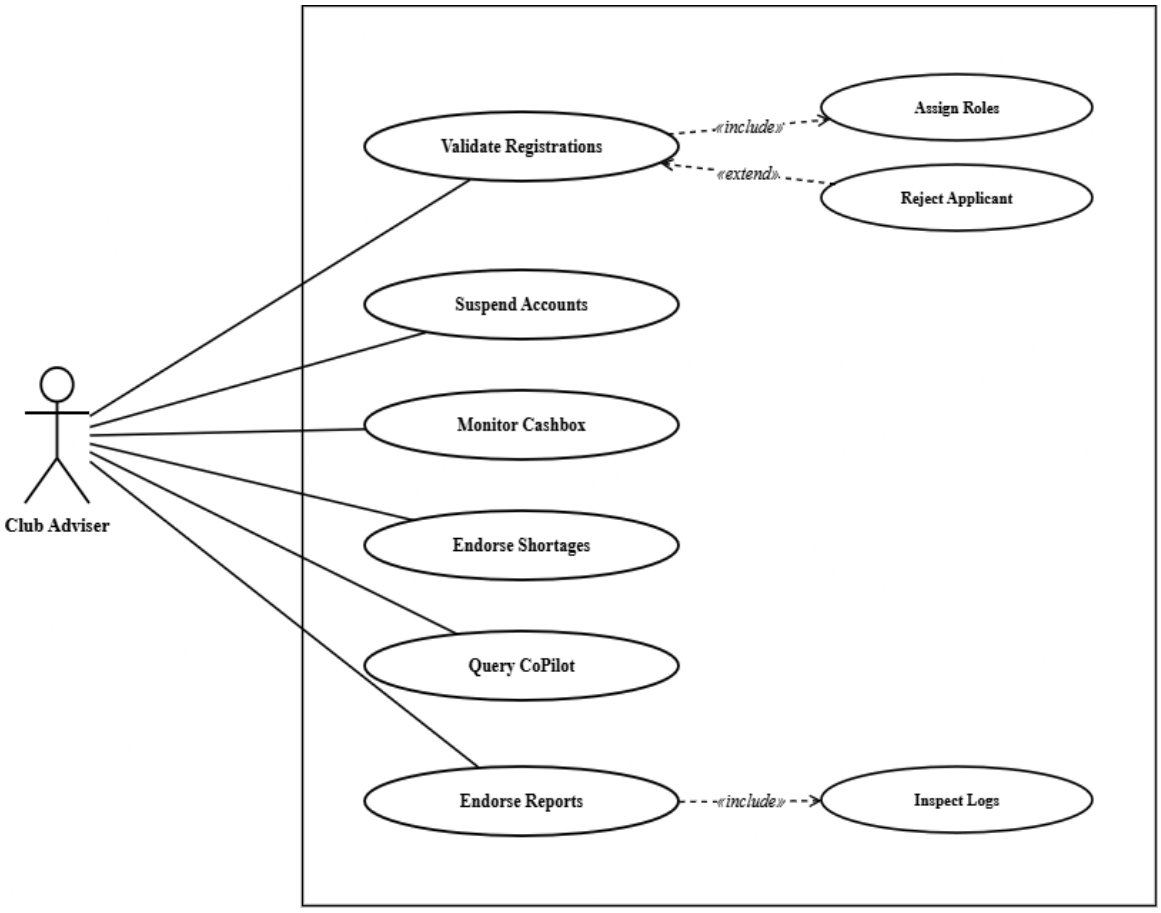


Figure 11: Use Case Diagram of Club Adviser

The Club Adviser acts as the system administrator and institutional authority, holding elevated governance privileges without participating in day-to-day cash handling. The Adviser initiates six base use cases within the system boundary: Validate Registrations, Suspend Accounts, Monitor Cashbox, Endorse Shortages, Query CoPilot, and Endorse Reports.

During user onboarding, the Adviser invokes Validate Registrations to verify applicant credentials, which mandatorily includes Assign Roles to attach role-based system permissions upon approval. If an applicant provides unauthorized or invalid credentials, the workflow branches into the conditional extension Reject Applicant to record the rejection reason and notify the applicant. For ongoing governance, the Adviser executes Suspend Accounts to revoke user access, Monitor Cashbox to observe physical cash totals, and Endorse Shortages to officially acknowledge cash discrepancies reported by the Treasurer. Through Query CoPilot, the Adviser asks the AI assistant natural-language budget questions to evaluate spending trends. Finally, when validating semester-end financial summaries, Endorse Reports includes Inspect Logs to verify the integrity of chronological audit trails before appending the final administrative endorsement.

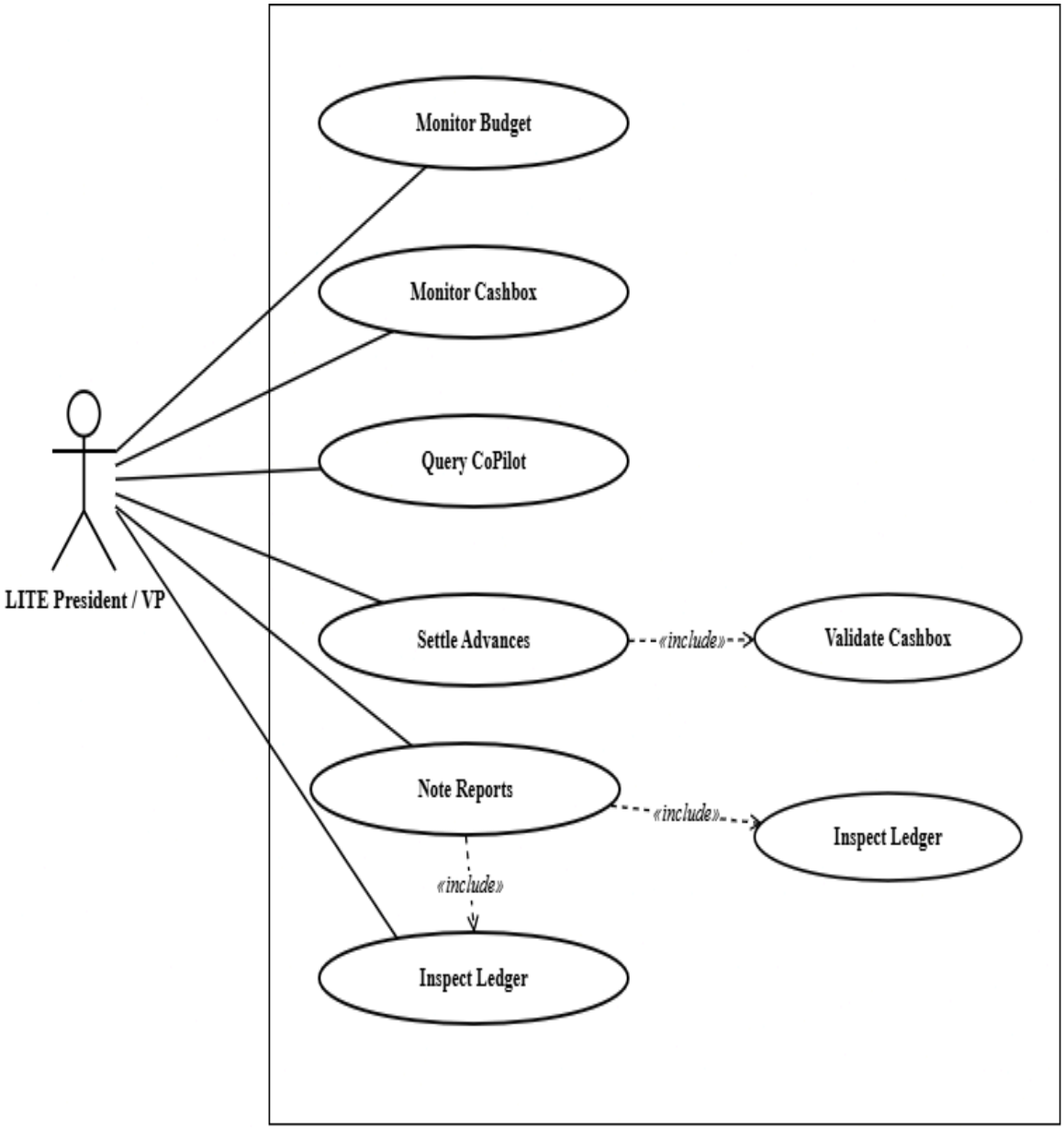


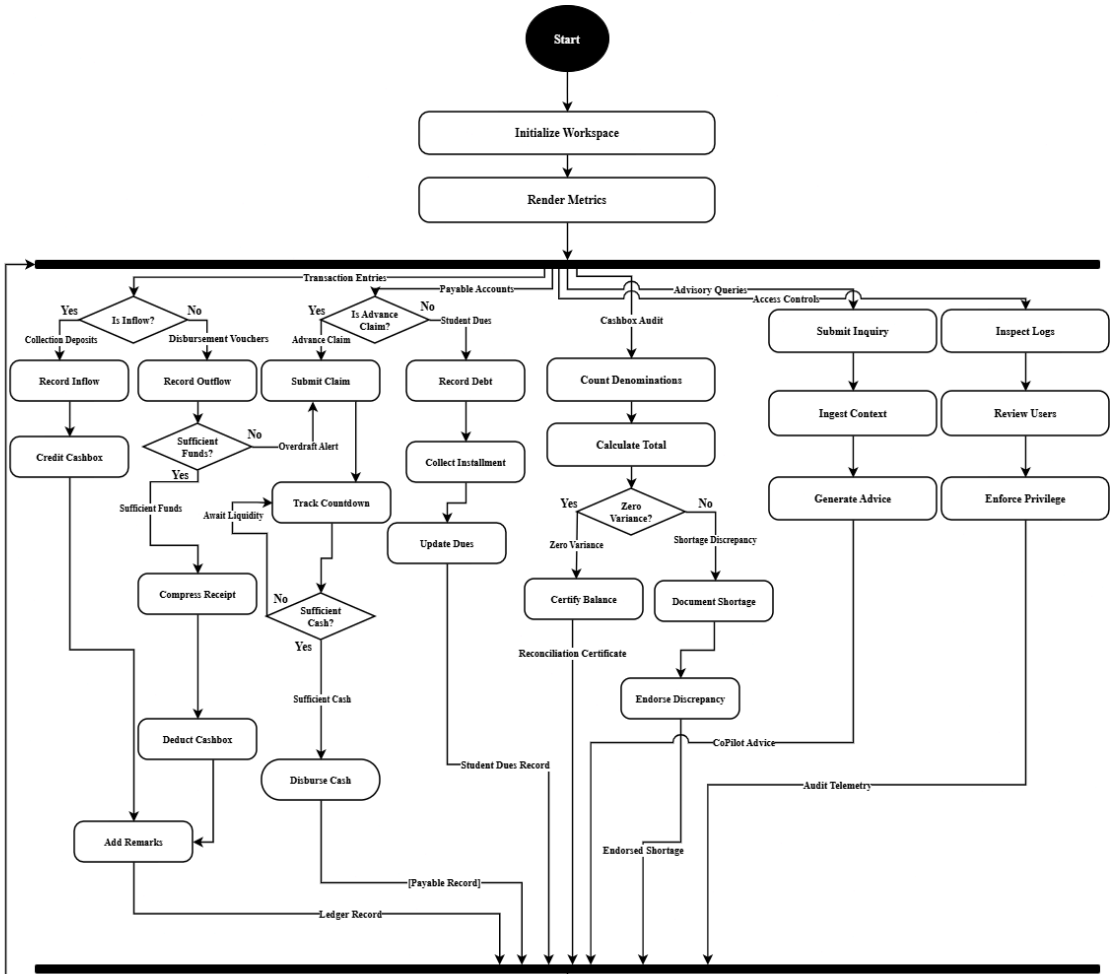
Figure 12: Use Case Diagram of LITE President/Vice President

The LITE President and Vice President comprise executive leadership, exercising programmatic steering, expenditure approvals, and official representation without handling daily physical cash. Executive officers directly access six base use cases: Monitor Budget, Monitor Cashbox, Query CoPilot, Inspect Ledger, Settle Advances, and Note Reports.

Through Monitor Budget and Monitor Cashbox, executive leadership observes high-level budget allocations, project expenses, and cash drawer liquidity. When

formulating project plans or evaluating burn rates, the officers invoke Query CoPilot to generate predictive summaries and category breakdowns from the AI engine. To inspect financial accountability, leadership executes Inspect Ledger to review itemized entries and supporting receipts. When reimbursing out-of-pocket expenses fronted by committee heads, Settle Advances includes the mandatory subroutine Validate Cashbox to confirm cash drawer solvency before approval. Finally, Note Reports includes Inspect Ledger to ensure executive leadership reviews ledger integrity before appending official notations to term-end reports submitted to the Dean.

2.1.4 Activity Diagram



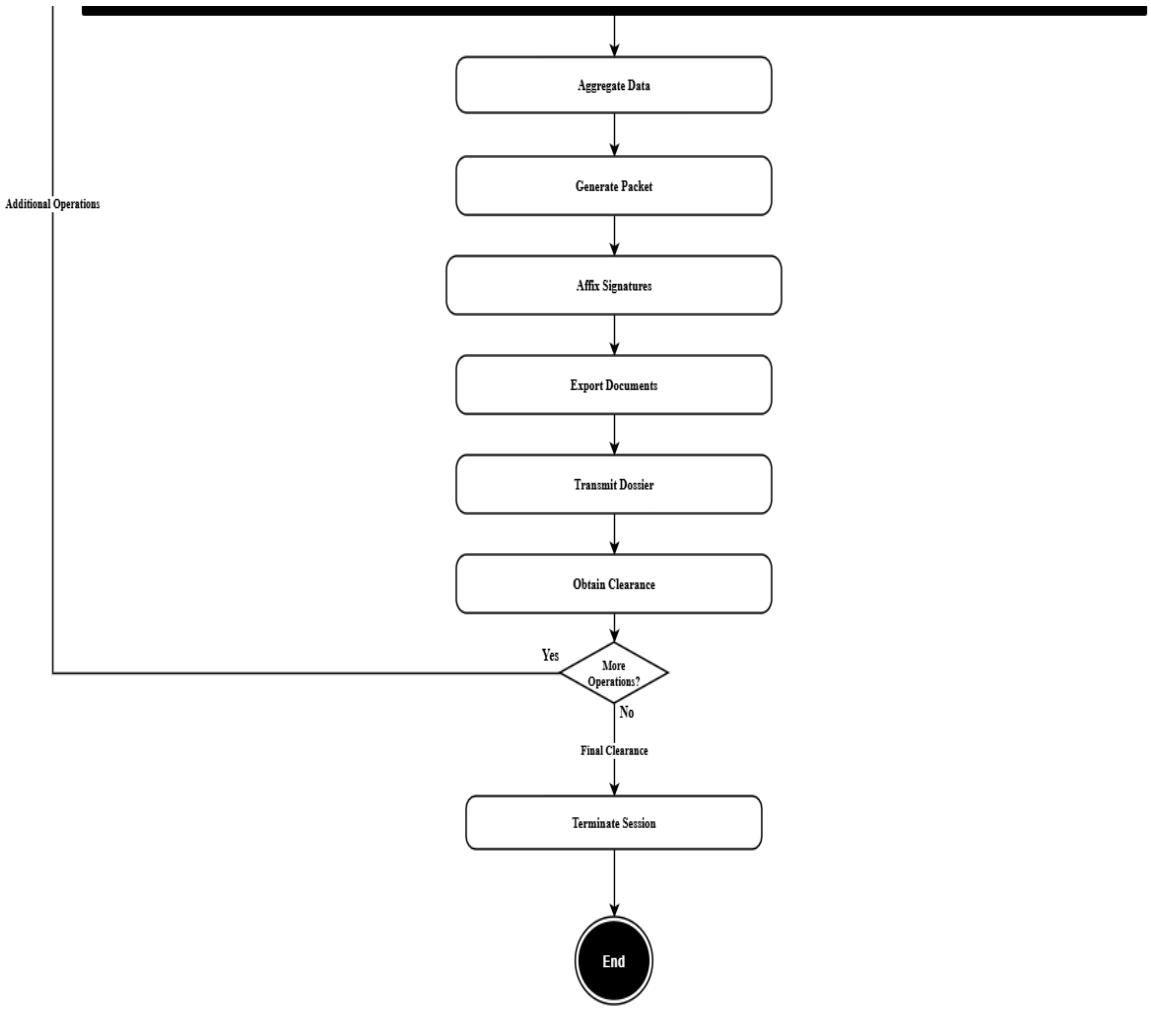


Figure 13: Activity Diagram of FinLITE System

The diagram starts at the initial node, where the system prepares the environment by running Initialize Workspace to load active semesters and account catalogs. It then runs Render Metrics to display real-time dashboard cards for cash on hand, pending advances, student receivables, and budget velocity. Control immediately moves into the solid black fork bar, which splits execution into five concurrent pipelines that run in parallel.

The first pipeline handles cash transactions by checking whether an entry is an inflow. If yes, then the system runs Record Inflow to capture collections and executes Credit Cashbox to add the cash. If no, then the entry is an expense, prompting the system to run Record Outflow and check whether there are sufficient funds. If yes, then it runs Compress Receipt to attach the voucher proof and executes Deduct Cashbox to subtract the money. If no, then it raises an overdraft alert and routes the expense into the payables

pipeline as a reimbursement claim. Both transaction paths then meet at Add Remarks to post internal notes before sending the ledger record to the join bar. In the second pipeline, the system checks whether an item is an advance claim. If yes, then an officer executes Submit Claim, tracks the seven-day liquidation countdown, and checks whether there is sufficient cash. If yes, then the treasurer executes Disburse Cash to release the reimbursement; if no, then the system waits for liquidity and holds the countdown until cash is available. If no on the advance claim check, then the item is student dues, directing the system to execute Record Debt, Collect Installment, and Update Dues before sending the balance to the join bar.

The remaining three pipelines handle physical cash auditing, intelligent assistance, and system governance. In the audit pipeline, the auditor counts all eleven Philippine coin and bill denominations, calculates the total, and checks whether there is zero variance. If yes, then the auditor executes Certify Balance to confirm that physical cash matches the books; if no, then the custodian executes Document Shortage and secures an adviser endorsement through Endorse Discrepancy. In the assistance pipeline, the user submits a prompt, allowing the system to ingest live balances and generate spending advice. In the governance pipeline, the system inspects audit logs, reviews user accounts, and enforces access privileges.

All five pipelines finish at the solid black join bar, which pauses execution until every concurrent stream is complete. Once synchronized, the system executes Aggregate Data, generates the standardized four-page liquidation packet, and collects digital approvals through Affix Signatures from the treasurer, auditor, president, and advisers. It then runs Export Documents to produce Word and PDF dossiers with receipt proofs, transmits the dossier to the program director and dean, and completes Obtain Clearance for formal semester clearance. Finally, the system checks whether more operations are needed. If yes, then it loops back up to the fork bar for additional work. If no, then it executes Terminate Session to log the user out, reaching the final end node.